

Legal Domain Quantitative Data: Seven Empires Compared

Infonomics Project — CAS Framework Domain Completeness Retrofit

Domain 2: Legal/Property Rights

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Executive Summary

The legal infrastructure of history's major empires exhibits remarkable quantitative variation, from the Tang Code's precisely enumerated **502 articles** to the Mongol Yasa's fragmented oral tradition with no surviving original text. This comprehensive data compilation reveals that codification levels, court system complexity, and sovereignty erosion patterns followed distinct trajectories across these seven polities, with Byzantine and Ottoman legal sovereignty loss to foreign merchants representing critical case studies for understanding how institutional weakness precedes collapse.

The research confirms significant quantitative data availability for Roman, Byzantine, Tang, and Ottoman legal systems, moderate availability for Venetian and Abbasid institutions, and fundamental evidentiary limitations for Mongol law. The most striking finding concerns legal sovereignty indicators: Byzantine commercial privileges to Italian merchants began in **1082** and culminated in complete sovereignty loss at the Fourth Crusade (**1204**), while Ottoman capitulations expanded from **1535** through the establishment of the Public Debt Administration (**1881**), which controlled **25-50%** of state revenues with a workforce exceeding the Finance Ministry itself.

I. ROMAN EMPIRE (27 BCE – 476 CE Western / 1453 CE Eastern)

A. Legal Code Metrics

Corpus Juris Civilis (529-565 CE) represents history's most influential legal codification. The **Digest** (promulgated December 16, 533 CE) contains **9,495 fragments** excerpted from **38-39 classical jurists**, condensing approximately **3 million lines** from 2,000 books into **150,000 lines** organized across **50 books** [HIGH confidence]. Ulpian alone contributed **33-40%** of all fragments. A commission of **16 members** led by Tribonian completed the compilation in **3 years**.

The **Codex Justinianus** (second edition November 16, 534 CE) comprises **12 books** containing approximately **4,562 imperial constitutions** spanning Hadrian to Justinian [HIGH confidence]. Book organization follows a logical structure: Book 1 covers ecclesiastical law and sources of law; Books 2-8 address private law; Book 9 treats criminal law; Books 10-12 contain administrative law. The first edition required only **14 months** of compilation by a **10-member commission**.

The **Institutes** (November 533 CE) provide a student textbook in **4 books**, with approximately **two-thirds** consisting of literal quotations from Gaius's Institutiones (~161 CE) [HIGH confidence]. Three compilers—Tribonian, Theophilus, and Dorotheus—produced this work simultaneously with the Digest.

The **Novellae Constitutiones** (534-565 CE) were never officially compiled during Justinian's lifetime. The Authenticum collection contains **134 novels**, while the later Greek Collection (575-580 CE) includes **168 novels** [HIGH confidence].

The earlier **Twelve Tables** (451-450 BCE) survive only in approximately **140 fragments** or paraphrases. Originally inscribed on **12 bronze tablets**, they were produced by **10 patrician commissioners** (decemviri) and ratified by the Centuriate Assembly in **449 BCE** [HIGH confidence].

Praetorian Edict codification occurred under Hadrian (~129 CE) through the work of Salvius Julianus. The reconstructed Edictum Perpetuum contains **5 parts, 45 titles, and 292 paragraphs** [MODERATE confidence—scholarly reconstruction].

Development	Date	Structure
Twelve Tables	451-450 BCE	12 tablets, ~140 surviving fragments
Edictum Perpetuum	~129 CE	5 parts, 45 titles, 292 paragraphs
Digest	533 CE	50 books, 9,495 fragments
Codex (2nd ed.)	534 CE	12 books, ~4,562 constitutions
Institutes	533 CE	4 books
Novellae	534-565 CE	134-168 novels (varying collections)

B. Court System Metrics

The number of **praetors** evolved significantly across Roman history [HIGH confidence]:

Period	Praetors
367 BCE (creation)	1
242 BCE	2
197 BCE	6
Sulla's reforms (~81 BCE)	8
Julius Caesar	10→16 (progressive increase)
Pomponius's time (2nd c. CE)	18

Quaestiones perpetuae (standing criminal courts) first appeared in **149 BCE** with the quaestio de repetundis for extortion. By Tacitus's time, approximately **16 standing courts** operated [HIGH confidence]. Juries typically contained **50-75 members**, with the jury album under Gracchan procedures listing **450 equestrian jurors**.

The transition from **legis actiones** (~450-130 BCE) through **formulary procedure** (~130 BCE-3rd century CE) to **cognitio extra ordinem** (early Principate onward) represents Roman procedural evolution. Formulary procedure was officially abolished in **342 CE** [HIGH confidence].

C. Property Rights Documentation

Roman property transfer distinguished between **res Mancipi** (requiring mancipatio with bronze, scales, and 5 witnesses) and **res nec Mancipi** (simple traditio) [HIGH confidence]. The **Lex Falcidia** (40 BCE) guaranteed heirs a minimum **1/4 of the estate**, capping legacies at **3/4**.

DATA GAP: Specific census enrollment figures and cadastral survey dates by province require specialized epigraphic research beyond available sources.

D. Contract Enforcement

Debt enforcement evolved from physical seizure (addictio) to the **Lex Poetelia Papiria** (327/326 BCE) abolishing debt bondage, then to **bonorum venditio** (~2nd century BCE) permitting sale of debtor's property en masse, and finally to **cessio bonorum** (Lex Julia, Caesar or Augustus period) allowing voluntary property surrender to avoid arrest and infamia [HIGH confidence].

E. Institutional Legal Development

The **Law School of Beirut** ("Nutrix Legum") operated under various dating estimates: Augustus period, Hadrianic reign (~130s CE), or Septimius Severus (~200 CE) [MODERATE confidence]. Under Justinian, **8 official law teachers** served the empire (**4 at Beirut, 4 at Constantinople**). The curriculum lasted **5 years**.

Documented student names total **51 from 20 provinces** [MODERATE confidence]. An earthquake and tsunami destroyed the school on **July 9, 551 CE**, with approximately **30,000 deaths**. The **Constantinople law school** was established under Theodosius II in **425 CE**.

The **Law of Citations** (426 CE) designated five authoritative jurists: Gaius, Papinian, Ulpian, Paulus, and Modestinus. Papinian's view prevailed in ties [HIGH confidence].

F. Legal Sovereignty Indicators

The **Constitutio Antoniniana** (212 CE) under Caracalla extended citizenship to virtually all free inhabitants, unifying legal jurisdiction [HIGH confidence]. Late empire legal fragmentation occurred through **episcopal audientia** (bishop's courts, from Constantine's reign) and expanded military jurisdiction.

DATA GAP: Quantitative data on episcopal court caseloads and jurisdictional scope changes require specialized research.

II. BYZANTINE EMPIRE (330-1453 CE)

A. Legal Code Metrics

Justinian's compilation initiated Byzantine legal tradition. Subsequent Greek-language codes include:

Ecloga (726/741 CE) under Leo III the Isaurian represented the first major Byzantine code in Greek rather than Latin, incorporating Christian principles and substituting mutilation for death in many cases [MODERATE confidence—exact chapter counts uncertain].

The **Basilika** (c. 892 CE), completed under Leo VI after initiation by Basil I, comprises **60 books** (τὸ ἐξήκοντάβιβλιον) subdivided into titles and chapters, representing a Greek translation/adaptation of Corpus Juris Civilis [HIGH confidence]. The critical edition by Scheltema et al. spans **17 volumes** (1955-1988). No complete manuscript survives.

The **Prochiron** (c. 870-879 CE) and **Epanagoge** (c. 879-886 CE) served as practical handbooks under Basil I.

The **Hexabiblos** (1344-1345 CE) by Constantine Harmenopoulos of Thessalonica contains **6 books** mirroring modern pandectist organization [HIGH confidence]. Approximately **70 manuscripts** survive. This work was adopted as the interim civil code of independent Greece in **1828** and remained influential through the 20th century in various jurisdictions.

Code	Date	Structure
Corpus Juris Civilis	529-534 CE	See Roman section
Ecloga	726/741 CE	Greek; abridged
Prochiron	c. 870-879 CE	Handbook format
Epanagoge	c. 879-886 CE	Expanded handbook
Basilika	c. 892 CE	60 books
Hexabiblos	1344-1345 CE	6 books

B. Court System Metrics

The **Eparch of Constantinople** served as urban prefect with supreme judicial jurisdiction, ranked just beneath the emperor. The **Kletorologion** of Philotheos (899 CE) lists **14 types** of subordinate officers under the Eparch [HIGH confidence]. The **Book of the Eparch** (c. 911-912 CE) regulated **21 distinct guilds** including **5 silk-related guilds** under special imperial scrutiny [HIGH confidence].

Constantine IX Monomachos created a ministry under the **epi ton kriseon** for civilian administrators in the **1040s**, and established a separate **nomophylax** position to supervise legal training.

C. Property Rights Documentation

The **Pronoia system** emerged during Constantine IX Monomachos's reign (1042-1055) and became the basis of the military system under the Komnenoi (1081-1185). Alexius I granted pronoia primarily to family members; Manuel I extended it to army officers in lieu of salary. Michael VIII Palaiologos made pronoia **hereditary after 1261** [HIGH confidence].

Praktika (imperial records) documented pronoia size, value, numbers of **paroikoi** (dependent peasants), and duties owed.

D. Contract Enforcement

The **Book of the Eparch** (c. 911-912 CE) specifies commercial regulations including profit margins: bakers were permitted **4% markup plus 16% for labor**; grocers could charge **up to 16%** [HIGH confidence]. The first guild chapter addresses **notary (tabelliones) regulations**.

Rhodian Sea Law, incorporated through Digest references, established maritime commercial principles including the general average principle for marine insurance [MODERATE confidence].

E. Institutional Legal Development

The original Constantinople law school (425 CE under Theodosius II) established **31 chairs total**: 10 Greek grammar, 10 Latin grammar, **2 law**, 1 philosophy, and 8 rhetoric [HIGH confidence]. Constantine IX Monomachos refounded the law school in **1045-1047 CE** with John Xiphilinos as first nomophylax. The requirement that school heads must know Latin persisted [HIGH confidence].

F. Legal Sovereignty Indicators (CRITICAL FOR THESIS)

Venice - Chrysobull Timeline:

Date	Emperor	Key Terms
1082	Alexios I Komnenos	Complete tax exemption throughout empire; merchant quarter in Constantinople; title of protosebastos for Doge
1126	John II	Reconfirmation
1147/1148	Manuel I	Explicit inclusion of Crete and Cyprus
1171	Manuel I	Mass arrest of Venetians; property confiscation (temporary)
1198	Alexios III Angelos	Extended privileges; mixed court provisions

The **1082 Chrysobull** granted Venice the "right to trade in all manner of merchandise in all parts of the empire free of any charge, tax, or duty payable to [the imperial] treasury." Historian Ostrogorsky assessed this as having "stifled the Empire's power of recuperation and ultimately started its terminal decline" [HIGH confidence].

Genoa received its first quarter at Galata/Pera in **1155** under Manuel I Komnenos. By **1352**, under John VI Kantakouzenos, Pera had become "un vero e proprio Stato entro lo Stato" (a true state within the state) with its own **Magnifica Communita** of **24 members** exercising civil, commercial, and administrative legal autonomy [MODERATE-HIGH confidence].

The **1182 Massacre of the Latins** killed an estimated **2,000-10,000** from a pre-massacre Latin population of approximately **60,000** in Constantinople (per Eustathius of Thessalonica). Approximately **4,000 survivors** were sold to the Turkish Sultanate of Rum [HIGH confidence].

The **Fourth Crusade** (1204) resulted in the sack of Constantinople (April 12-15, 1204) with total loot estimated at **~900,000 silver marks**. Venice received **3/8 of the Byzantine Empire** in territorial claims. Byzantine law was suspended in Latin territories [HIGH confidence].

The **Treaty of Nymphaeum** (1261) granted Genoa access to the Black Sea and extensive commercial privileges in exchange for assistance (ultimately not required) in recapturing Constantinople [HIGH confidence].

Sovereignty Erosion Timeline:

Year	Event	Impact
1082	Venice Chrysobull	Tax exemption begins revenue loss
1111	Pisa privileges	Second Italian power gains foothold
1155	Genoa privileges	Third major competitor
1204	Fourth Crusade	Complete sovereignty loss
1261	Treaty of Nymphaeum	Sovereignty partially restored but commercially compromised
1267-1304	Galata/Pera development	Semi-autonomous Genoese enclave
1453	Fall of Constantinople	Final sovereignty extinction

This represents approximately **370 years of progressive legal sovereignty erosion** from 1082 to 1453.

III. TANG DYNASTY CHINA (618-907 CE)

A. Legal Code Metrics (BENCHMARK DOCUMENT)

The **Tang Code** (**Tang lü shuyi**, **653 CE final version**) contains **502 articles** organized into **12 sections (chapters)** across **30 volumes (juan)** [HIGH confidence—this is the benchmark for comparative analysis]. The General Principles section comprises **57 articles** (Chapters I-VI), representing nearly one-third of total text. The Specific Articles section contains **445 articles** (Chapters VII-XXX). Three articles (223, 300, 502) define concepts rather than prescribing punishments.

12 Chapter Structure:

- 1. **Mingli** (名例) - Terms and Examples/General Definitions
- 2. **Weijin** (衛禁) - Security and Prohibitions
- 3. **Zhizhi** (職制) - Administrative Regulations
- 4. **Huhun** (戶婚) - Household and Marriage
- 5. **Jiuku** (廐庫) - Public Stables and Warehouses

- 6. **Shanxing** (擅興) - Unauthorized Levies
- 7. **Zeidao** (賊盜) - Violence and Robbery
- 8. **Dousong** (鬥訟) - Assaults and Accusations
- 9. **Zhawei** (詐偽) - Fraud and Counterfeit
- 10. **Zalu** (雜律) - Miscellaneous Articles
- 11. **Buwang** (捕亡) - Arrest and Flight
- 12. **Duanyu** (斷獄) - Judgment and Imprisonment

The **commentary (shuyi)** was compiled 652-653 CE in question-and-answer format, often equaling or exceeding the length of the law articles themselves [HIGH confidence].

Code Development Timeline:

Date	Code	Status
581 CE	Kaihuang lü (Sui)	Predecessor basis
624 CE	Wude lü	First Tang code
637/639 CE	Zhenguan lü	Lightened punishments
651 CE	Yonghui lü	Modernized version
653 CE	Tanglü shuyi	Final version with commentary

The **Tang Liudian** (Six Codes, 738-739 CE), commissioned by Emperor Xuanzong in **722 CE** and compiled by Li Linfu, represents China's earliest complete administrative law code, organized by the Six Ministries and 24 divisions [HIGH confidence].

B. Five Punishments Specifications

Punishment	Chinese	Grades
Light Beating (Chi 笞)	10-50 strokes	5 grades: 10, 20, 30, 40, 50
Heavy Beating (Zhang 杖)	60-100 strokes	5 grades: 60, 70, 80, 90, 100
Penal Servitude (Tu 徒)	1-3 years	5 grades: 1, 1.5, 2, 2.5, 3 years
Exile (Liu 流)	Distance from capital	3 grades: 2,000, 2,500, 3,000 li

Punishment	Chinese	Grades
Death (Si 死)	Strangulation or Decapitation	2 forms

144 crimes were punishable by strangulation; **89 crimes** by decapitation [MODERATE confidence]. Execution was restricted to approximately **60 days per year** due to seasonal and religious prohibitions.

C. Court System Metrics

Three Central Judicial Organs:

The **Court of Judicial Review (Dalisi)** served as the supreme criminal court. Staff structure included a Chief Minister (Rank 3B), 2 Vice Ministers (Rank 5B), 2 Directors, and **100 inquisitors (wenshi)** [HIGH confidence —based on institutional records].

The **Ministry of Justice (Xingbu)** reviewed exile and death penalty verdicts from Dalisi. The **Censorate (Yushitai)** performed supervisory/investigative judicial functions.

Torture regulations specified a maximum of **200 blows total**, with a maximum of **3 interrogation sessions** and a minimum **20-day interval** between sessions. The permitted interrogation stick measured approximately **40 inches (101.6 cm)** in length.

D. Property Rights: Equal-Field System (Juntian)

The **Tang Dynasty Land Allocation** (624 CE Wude Decree) specified [HIGH confidence]:

Category	Allocation
Adult male (ding)	100 mu total
— Koufen tian (returnable)	80 mu
— Yongye tian (heritable)	20 mu
Infirm/disabled	40 mu
Widows/concubines	30 mu

System breakdown indicators appeared by the late 7th century. The **Two-Tax System** formally replaced the equal-field system in **780 CE** [HIGH confidence].

E. Census Data

Year	Households	Population
742 CE	~9 million	~50 million
754 CE	—	1,859 cities, 321 prefectures, 1,538 counties

Chang'an contained approximately **2 million people** with the city proper housing **~1 million**—the world's largest at the time. Military troops numbered approximately **500,000** by 742 CE [HIGH confidence].

F. Examination System

Jinshi degrees awarded (entire Tang dynasty): 6,585 total, averaging approximately **23 per year** with a pass rate of **1-2%** from **1,000-2,000 annual candidates** [HIGH confidence].

G. International Legal Transmission

The **Japanese Taihō Ritsuryō** (701 CE) comprised **6 volumes of penal law (ritsu)** and **11 volumes of administrative law (ryō)**, directly based on Tang's Yonghui Code (651 CE). A commission of **18 Japanese aristocrats plus 1 Chinese scholar** produced the compilation. The ritsu followed the Tang criminal code "almost as it was" [HIGH confidence].

IV. ABBASID CALIPHATE (750-1258 CE)

A. Structural Note on Islamic Jurisprudence

FUNDAMENTAL DIFFERENCE: Islamic law (fiqh) is NOT a code promulgated by legislative authority but rather a **jurisprudential tradition** derived through scholarly interpretation (ijtihad) of Quran and Sunnah. Multiple valid interpretations coexist within orthodox frameworks. This creates methodological challenges for quantification—confidence levels are necessarily **MODERATE-LOW** for numerical metrics.

B. Four Sunni Madhabs

School	Founder	Death Date	Abbasid-Era Distribution
Hanafi	Abu Hanifa al-Nu'man	767 CE	Iraq, Persia, Central Asia; official Abbasid madhab
Maliki	Malik ibn Anas	795 CE	North Africa, al-Andalus
Shafi'i	al-Shafi'i	820 CE	East Africa, Southeast Asia, parts of Arabia

School	Founder	Death Date	Abbasid-Era Distribution
Hanbali	Ahmad ibn Hanbal	855 CE	Arabia, parts of Iraq; smallest geographical spread

Consolidation into four schools occurred by the 9th-10th centuries. By the late 3rd/9th century, "hardly any scholars adhered to any other approach" [HIGH confidence for dates; MODERATE for geographic spread].

C. Major Legal Treatises

Al-Muwatta (Malik ibn Anas, completed c. 798 CE) required approximately **40 years** of composition. The Yahya al-Laythi transmission contains **1,942 narrations**; the Abu Mus'ab al-Zuhri transmission includes **3,069 items**. Structure: **61 books (kitab)** with internal chapters. Malik reportedly selected these ~1,900 narrations from approximately **100,000** available. **Over 1,000 disciples** transmitted the work, with **16 documented versions** and approximately **100 commentaries** on the Yahya version alone [HIGH confidence for structure; MODERATE for precise counts].

Kitab al-Umm (al-Shafi'i, post-814 CE) spans **4-10 volumes** depending on edition. Al-Shafi'i authored "more than 100 books" according to sources [MODERATE confidence].

D. Court System Metrics

Qadi al-Qudat (Chief Qadi) position was established **786-798 CE under Harun al-Rashid**, with **Abu Yusuf** (d. 798 CE) as first holder [HIGH confidence].

Mazalim Courts (grievance tribunals) trace to Abd al-Malik ibn Marwan (r. 685-705 CE), who first assigned a day for deliberation of wrongs. Umar II (r. 717-720 CE) instituted a formal office. Al-Mawardi (d. 1058) systematized **10 areas of jurisdiction** [HIGH confidence].

Early Abbasid qadi courts remained "rudimentary" with "few judicial auxiliaries" in the 750s. Professional witnesses (shuhud) emerged to compensate for foreign qadis' lack of local connections.

E. Property Rights

Waqf system documentation includes the oldest dated waqfiya (deed) from **876 CE** (Turkish and Islamic Arts Museum, Istanbul). The earliest known Egyptian waqf dates to **919 CE** [HIGH confidence].

Iqta system underwent major systematization during the Buyid period (945-1055). Mu'izz al-Dawla (945-946) began systematic allocation of caliphal estates to soldiers. Iqta represented revenue assignment, NOT land ownership—theoretically non-hereditary and revocable [HIGH confidence].

F. Institutional Development

The **Nizamiyya Baghdad** madrasa was founded **1065-1067 CE** by Nizam al-Mulk. Al-Ghazali served as professor **1091-1095 CE** (appointed at age 33). The institution was destroyed in the Mongol sack of Baghdad in

1258 CE. Student enrollment estimates reach up to **3,000** by the late 11th century [SPECULATIVE for enrollment; HIGH for dates].

Nizam al-Mulk established a network of Nizamiyya madrasas in Nishapur, Isfahan, Herat, Balkh, and Amol during **1063-1092 CE** [HIGH confidence].

G. Legal Sovereignty Indicators

Buyid capture of Baghdad (945 CE) severely diminished caliphal legal authority. The caliph retained only religious/symbolic authority while sultans exercised de facto power. Waqf institutions created economic spheres partially independent of state control.

DATA GAP: Numbers of qadis per city, case load statistics, and contract enforcement rates unavailable due to non-preservation of administrative records and the nature of fiqh as distributed jurisprudential tradition rather than codified law.

V. OTTOMAN EMPIRE (1299-1922 CE)

A. Legal Code Metrics

The Ottoman system maintained **dual legal structures**: Sharia courts (Hanafi school official) and Kanun (sultanic/secular law). Ebussuud Efendi served as Şeyhülislam **1545-1574**, aligning kanun with sharia [HIGH confidence].

Mehmed II's Kanunnames (1444-1481) comprised three compilations: (1) conditions and obligations of subjects (1453-1456); (2) organization of state and Ruling Class (1477-1478); (3) economic organization, landholding, taxes. The kanun crystallized into final form by **1501** [HIGH confidence].

Suleiman's Kanunname-i Al-i Osman compiled all previous kanunnames into a single sultanic code lasting over **300 years**. Suleiman earned the title "Kanuni" (The Lawgiver) [HIGH confidence].

DATA GAP: Exact article counts for Mehmed II's and Suleiman's kanunnames not found in available sources.

B. Tanzimat Legal Reforms (1839-1876)

Year	Code	Model/Structure
1850	Commercial Code	French model
1858	Ottoman Penal Code	French Napoleonic model
1858	Land Code	132 articles in 3 parts

Year	Code	Model/Structure
1864	Vilayet Law	French provincial model
1869-1876	Mecelle	1,851 articles across 16 books

The **Mecelle** (1869-1876) contains **1,851 articles** organized into **16 books** with **99 legal maxims** guiding judicial interpretation [HIGH confidence—VERIFIED]. Commission Chair: Ahmed Cevdet Pasha. The Mecelle remained in force in Turkey until **1926**, Lebanon until **1932**, Syria until **1949**, Iraq until **1953**, and continues as the basis of civil law in Jordan and Kuwait.

C. Court System Metrics

Traditional structure comprised three systems: Muslim courts (Hanafi), non-Muslim millet courts, and trade courts. The Şeyhülislam headed the ulema hierarchy, with Kazaskers (military judges) for Rumelia and Anatolia below.

Nizamiye Courts (secular courts for new codes) were established in **1864**, modeled on the French three-tiered system. The **Mekteb-i Nüvvab** (Naibs' College, founded **1855**) and **Mekteb-i Hukuk** (Law School, **1878**) trained judges [HIGH confidence].

D. Property Rights

Tahrir defterleri (cadastral surveys) first appear in **1431** (Albania, reign of Murad II). Total land units registered reached approximately **60 million** (per Bıyık and Yomralıoğlu 1994). Legal survey frequency: every **30-40 years** [HIGH confidence].

The **Land Code of 1858** established **5 land categories**: Mülk (private freehold), Miri (state-owned, cultivator usufruct), Waqf (religious endowment), Matruke (public lands), and Mawat (wasteland). Abandonment of miri cultivation for **3 years** meant reversion to state [HIGH confidence].

E. Legal Sovereignty Indicators (CRITICAL FOR THESIS)

Capitulations Timeline:

Year	Nation	Key Terms
1535/1536	France	First major capitulation; extraterritorial consular jurisdiction; 3-5% customs
1580	England	Equal footing with French
1612	Dutch	Extended capitulations
1740	France	"Pivotal agreement" - 85 articles ; model for subsequent treaties
1774	Russia	Treaty of Küçük Kaynarca - CRITICAL (see below)
1830	United States	American capitulatory privileges
1914	—	Ottoman unilateral abrogation
1923	—	Treaty of Lausanne Article 28: complete abolition

Treaty of Küçük Kaynarca (July 21, 1774) granted Russia the right to build an Orthodox church in Constantinople and make "representations" concerning Orthodox Christians. Russia interpreted this as protectorate rights over ALL Orthodox Christians in the Ottoman Empire. War reparations: **4.5 million rubles**. This created the legal basis for frequent Russian interventions for **80+ years** [HIGH confidence].

Ottoman Public Debt Administration (December 20, 1881):

- Debt before restructuring: **£191 million**
- Debt after restructuring: **£106 million** (~40% reduction)
- Yearly charges reduced by: **>80%**
- Governing Council: **7 members** (British, French, German, Austrian, Italian, Dutch creditors + 1 Ottoman)
- Employees: **5,000-9,000 officials** (exceeded Ottoman Finance Ministry)
- Revenue Control: **25-50% of state revenues**

OPDA controlled: salt monopoly, tobacco monopoly (Régie des Tabacs), stamp taxes, alcohol taxes, fishing taxes, silk tithe, Bulgaria tribute, Cyprus surplus, Eastern Rumelia revenues, and customs duties. Young Turks viewed OPDA as a direct threat to independence [HIGH confidence].

Foreign Post Offices: Austria operated **65 locations** across the Balkans. Ottoman domestic post offices numbered only **63** throughout the entire empire in **1863** [HIGH confidence].

Ali Pasha's 1856 Critique: Extraterritorial jurisdiction created "a multiplicity of governments in the Government."

VI. VENETIAN REPUBLIC (697-1797 CE)

A. Legal Code Metrics

Promissione Ducale evolved from approximately **12 pages** (1192, Enrico Dandolo) to **300+ pages** (1797, Ludovico Manin) [HIGH confidence]. **Correttori** (5 "correctors of the promissione ducale") were established in **1229**.

Jacopo Tiepolo's Statutes (1242) comprised **5 books** of civil statutes codifying civil and economic relations. A **sixth book** was added in **1346** by Doge Andrea Dandolo [HIGH confidence].

The **Venetian Patent Statute** (March 19, 1474) represents the **first codified patent law in the world** [HIGH confidence].

B. Court System Metrics

Court	Established	Jurisdiction
Quarantia Criminale	1179-1223	Criminal law
Quarantia Civil Vecchia	1441	Civil law, Venice/Stato da Mar
Quarantia Civil Nuova	1491/1492	Civil law, terraferma

Each Quarantia contained **40 judges** elected from the Great Council. Total judges: **120** (plus additional panels totaling **160**) [HIGH confidence].

The **Council of Ten** was created **July 15, 1310** in response to the Bajamonte Tiepolo conspiracy. Composition: **17 members** (10 elected patricians + Doge + 6 councillors). **State Inquisitors** (3 judges from Ten) were established in **1539** [HIGH confidence].

C. Great Council Metrics

Date	Membership
1172	~35 initial
1297 (Serrata)	586
1340	1,212
16th century typical	2,095

Date	Membership
Maximum recorded	2,746

The **Golden Book (Libro d'Oro)** registry was created in **1315**; birth records established **1506**; marriage records **1526** [HIGH confidence].

D. Contract Enforcement

Colleganza contracts first documented in **1073**; hundreds survive in the Puga and Trefler database covering 1073-1342 [HIGH confidence]. Standard structure: **Stans** (sedentary investor) provided 100% capital for 75% profits; **Tractor** (traveling merchant) provided 0% capital for 25% profits. The colleganza introduced **limited liability**—a major innovation over Roman law.

Banco della Piazza di Rialto (first public bank) was established **1584/1587**. Peak deposits reached **1.7 million ducats** with approximately **4,000 depositors** in **1618**. The **Banco del Giro** (established **1619**) represented "likely the first experiment with purely fiduciary state-issued legal-tender money" [HIGH confidence].

E. Legal Sovereignty Indicators

Venice as Beneficiary of Byzantine Sovereignty:

Date	Acquisition
1082	Chrysobull: tax-free trade throughout Byzantine Empire
1204	Fourth Crusade: 3/8 of Byzantine Empire territorial claim
1211-1669	Crete (458 years)

Doge's title post-1204: "**Dominator quartae partis et dimidiaie totius Imperii Romaniae**" (Lord of One-Quarter and One-Eighth of the Entire Byzantine Empire) [HIGH confidence].

Venice State Archive preserves documents from **976 CE** onward with approximately **~80 km of shelving** and **800+ fonds** [HIGH confidence].

End of Sovereignty: Great Council abdication **May 12, 1797**; **Treaty of Campo Formio** (October 17, 1797) partitioned Venice between Austria and France.

VII. MONGOL EMPIRE (1206-1368 CE)

A. Critical Evidentiary Limitations

NO original Mongol text of the Great Yasa survives—not a single authenticated Mongol-language scroll or codex has been found. All reconstructions derive from fragmentary secondary sources in Arabic, Persian, Armenian, Byzantine, and Western European languages. David Morgan (*BSOAS* 1986) demonstrated "the lack of evidence for the written code" and questioned whether a comprehensive codified law ever existed [HIGH confidence for the limitation itself].

B. Yasa Reconstructions

François Pétis de la Croix (1710) compiled approximately **36 provisions** from Persian and Arabic chroniclers [LOW confidence for content; HIGH confidence that reconstruction exists].

Primary Sources with Limitations:

Source	Date	Limitation
Juvaini	c. 1260	Never saw scrolls; describes "rambling" principles
Rashid al-Din	c. 1310	Lacks specifics
al-Maqrizi	15th c.	Derived from al-Umari

Death Penalty Provisions (cross-source attestation): adultery, theft (third offense), spying, witchcraft, urinating in water, military desertion, failure to assist fellow soldiers [MODERATE confidence].

Religious Tolerance: Consistently attested; corroborated by documentary evidence including Golden Horde yarlyks granting Russian Church tax immunity [HIGH confidence].

C. Yarghuchi System

Shigi Qutuqu was appointed first yarghuchi at the **1206** founding kurultai. The Secret History §203 records Chinggis Khan's instruction to "write the laws in a blue book" [HIGH confidence for appointment; LOW for what was actually written].

Ilkhanate evidence: Three yarghuchi appointment charter samples (yarliks) preserved in the Dastūr al-kātib (1360s) detail appointment preconditions [MODERATE-HIGH confidence].

D. Property Rights

Appanage allocations began in **1207** to Chinggis Khan's sons and brothers. Property rights structure: Khan held ultimate sovereignty; princes/princesses held "proprietary rights" over ordos and resources; appanages

were heritable [MODERATE confidence].

Ortaq (merchant partnership) system institutionalized under Chinggis Khan. Contracts "closely resembled qirad and commenda arrangements" from Islamic and Mediterranean commercial law. Merchants received access to the yam system, tax exemptions, low-interest loans, and the right to bear arms [MODERATE-HIGH confidence].

Paiza (safe-conduct tablet) system: Gold for highest elites; silver for mid-level officials; iron with silver inlay for messengers. **Ghazan Khan reforms** (1295-1304): paizas bear bearer names, graded seals, **30-year expiration**, must be surrendered at term end. Physical artifacts survive (Met Museum 1993.256) [HIGH confidence].

E. Yuan Legal Codes

Zhiyuan xinge (1291): First Yuan codification attempt; incomplete.

Da Yuan tongzhi (1323): **88 juan**; modeled after Jin dynasty Taihe codes. Only Tongzhi tiaoge survives fragmentarily as **22 juan** Ming manuscript.

Zhizheng tiaoge (1346): Last Yuan "legal code"; approximately **1.5x length** of Da Yuan tongzhi. Less than half discovered in **2003** in a private library in Gyeongju, South Korea; designated South Korean **National Treasure (February 2021)** [HIGH confidence for discovery].

Key finding: "The Yuan dynasty never produced a Chinese-style legal code. Instead, it compiled imperial edicts and judicial decisions for local administration" [HIGH confidence — scholarly consensus].

F. Legal Sovereignty Indicators

Subject Population Autonomy:

Territory	Autonomy Level
Yuan China	MODERATE (Chinese procedures continued)
Ilkhanate pre-1295	MODERATE (yasa vs. sharia tension)
Ilkhanate post-1295	HIGH for Muslims (sharia dominant after Ghazan's conversion)
Golden Horde	HIGH (baskaki withdrawn 1328)

Golden Horde yarlyks: **6 authentic documents** preserved; **1 16th-century forgery** identified. Russian Orthodox Church received tax immunity [HIGH confidence].

VIII. CROSS-EMPIRE COMPARISON TABLES

Table 1: Legal Code Structure Comparison

Empire	Primary Code	Date	Article/Unit Count	Confidence
Roman	Corpus Juris Civilis (Digest)	533 CE	9,495 fragments (50 books)	HIGH
Byzantine	Basilika	c. 892 CE	60 books	HIGH
Tang	Tang Code	653 CE	502 articles (12 chapters)	HIGH
Abbasid	<i>No unified code</i>	—	Madhab-based jurisprudence	N/A
Ottoman	Mecelle	1869-1876	1,851 articles (16 books)	HIGH
Venetian	Tiepolo Statutes	1242	5 books (+1 in 1346)	HIGH
Mongol	Great Yasa	1206?	~36 reconstructed provisions	SPECULATIVE

Table 2: Court System Complexity

Empire	Hierarchy Levels	Specialized Courts	Key Metrics
Roman	4+ (local→praetor→quaestiones→emperor)	~16 quaestiones perpetuae	50-75 jurors per court
Byzantine	4+ (local→provincial→eparch→emperor)	Commercial, ecclesiastical	21 regulated guilds
Tang	4 (county→prefecture→central→emperor)	Dalisi, Xingbu, Yushitai	100 inquisitors at Dalisi
Abbasid	3 (qadi→qadi al-qudat→caliph)	Mazalim, Hisba	10 mazalim jurisdiction areas
Ottoman	3 (qadi→kazasker→şeyhülislam)	Nizamiye (1864), mixed	Multiple parallel systems
Venetian	3-4 (local→Quarantia→Council of Ten)	3 Quarantia divisions	120-160 total judges
Mongol	2-3 (local→yarghuchi→khan)	Subject populations retained own	Limited institutionalization

Table 3: Legal Education Institutions

Empire	Primary Institution	Founded	Key Metrics
Roman	Law School of Beirut	~130s-200s CE	51 documented students; 5-year curriculum
Byzantine	Constantinople Law School	425 CE / refounded 1045 CE	31 chairs (425 CE); 2 for law
Tang	Imperial University (Guozijian)	Sui/Tang	23 jinshi/year average; 1-2% pass rate
Abbasid	Nizamiyya Baghdad	1065-1067 CE	Est. ~3,000 students (speculative)
Ottoman	Mekteb-i Hukuk	1878	Western-style legal training
Venetian	Padua University	Under Venetian control 1405-1797	Law faculty active
Mongol	No formal legal schools	—	Apprenticeship/oral tradition

Table 4: Legal Sovereignty Loss Indicators (CRITICAL)

Empire	Key Sovereignty Erosion	Date	Mechanism	Final Loss
Roman	Provincial/religious fragmentation	3rd-5th c.	Episcopal audientia; military courts	476 CE (West)
Byzantine	Italian merchant privileges	1082-1204	Chrysobulls; tax exemption	1204 (Fourth Crusade)
Tang	Jiedushi autonomy	755 CE (An Lushan)	Regional military governors	907 CE
Abbasid	Buyid capture of Baghdad	945 CE	Military dynasties; caliphal authority nominal	1258 CE
Ottoman	Capitulations→OPDA	1535-1881	Extraterritorial jurisdiction; debt control	1922 CE
Venetian	Campo Formio	1797	Napoleonic conquest	1797 CE
Mongol	Khanate fragmentation	1260	Civil war; succession disputes	1368 CE (Yuan)

Table 5: Property Rights Documentation Systems

Empire	Land Registration	Cadastral Survey Dates	Key Features
Roman	Provincial/Italian distinction	Augustan period	Mancipatio/traditio distinction
Byzantine	Imperial records (praktika)	Macedonian dynasty (10th c.)	Pronoia documentation
Tang	Census/tax registration	742 CE major census	Equal-field system (100 mu/adult male)
Abbasid	Tax cadastres (dafatir)	Various	Waqf deeds from 876 CE; iqta system
Ottoman	Tahrir defterleri	From 1431; every 30-40 years	~60 million land units registered
Venetian	State archives	From 976 CE	~80 km shelving; notarial records
Mongol	Limited formal registration	—	Appanage system; ortaq contracts

IX. DATA GAPS SUMMARY

Roman Empire

- Precise census enrollment figures by province
- Detailed court caseload statistics
- Provincial cadastral survey dates

Byzantine Empire

- Ecloga exact chapter count
- Specific customs revenue figures (quantitative form)
- Provincial court numbers

Tang Dynasty

- Student counts at Imperial University law school
- Specific article counts per chapter of Tang Code
- Case volume statistics

Abbasid Caliphate

- Numbers of qadis per city
- Case load statistics
- Contract enforcement rates
- Precise chronology of professional witness institutionalization

Ottoman Empire

- Kanunname article counts (Mehmed II, Suleiman)
- Total sharia court numbers across empire
- Provincial kanunname totals

Venetian Republic

- Senate decree volumes
- Maritime law (Capitolare Nauticum) structure
- Case volumes

Mongol Empire (MOST SIGNIFICANT GAPS)

- No original Yasa text
- Yarghuchi appointment numbers
- Case statistics
- Property registration systems

X. CONFIDENCE LEVEL SUMMARY

Empire	Overall Data Quality	Quantitative Reliability
Roman	HIGH	Excellent for Corpus Juris Civilis; gaps in administrative data
Byzantine	HIGH-MODERATE	Strong for legal codes; moderate for institutional metrics
Tang	HIGH	Benchmark precision for code structure

Empire	Overall Data Quality	Quantitative Reliability
Abbasid	MODERATE-LOW	Structural difference (fiqh) limits quantification
Ottoman	HIGH	Excellent for Tanzimat era; gaps for kanun period
Venetian	HIGH	Exceptional archival preservation
Mongol	LOW-SPECULATIVE	Fundamental evidentiary limitations

XI. KEY FINDINGS FOR CAS FRAMEWORK FORMALIZATION

Legal Sovereignty Erosion as Collapse Precursor: The data supports examining legal sovereignty loss as an indicator preceding collapse:

- **Byzantine:** 1082 Chrysobull → 1204 Fourth Crusade (**122 years** of progressive erosion)
- **Ottoman:** 1535 Capitulations → 1881 OPDA → 1922 collapse (**387 years** with accelerating sovereignty loss)
- **Tang:** 755 An Lushan → 907 collapse (**152 years** of jiedushi autonomy)
- **Abbasid:** 945 Buyid capture → 1258 Mongol conquest (**313 years** of nominal caliphal authority)

Quantifiable Metrics for Modeling:

1. **Code article counts** (Tang 502 vs. Mecelle 1,851 vs. Digest 9,495 fragments)
2. **Court hierarchy levels** (range: 2-4+ across empires)
3. **Cadastral survey frequency** (Ottoman: 30-40 years)
4. **Foreign privilege expansion dates** (trackable for Byzantine and Ottoman)
5. **Revenue control percentages** (OPDA: 25-50%)

Structural Variation: Islamic jurisprudential traditions (Abbasid, Ottoman sharia component) resist direct quantification in ways that Roman-derived codified systems do not. Mathematical modeling should account for this structural difference.

Benchmark: The Tang Code's **502 articles across 12 chapters** provides the clearest comparative anchor for measuring codification precision across legal traditions.

SCHOLARLY SOURCES REFERENCED

Roman Law: Wolfgang Kaiser, Bruce Frier, Alan Watson, Tony Honoré, David Johnston, Jill Harries, Cambridge Companion to Roman Law

Byzantine Law: Bernard Stolte, Spyros Troianos, Ruth Macrides, Angeliki Laiou, H.J. Scheltema (Basilicorum libri LX critical edition), David Jacoby

Tang Law: Wallace Johnson (Tang Code translation), Denis Twitchett (Cambridge History of China), Karl Büniger, Jacques Gernet

Abbasid Law: Joseph Schacht, Wael Hallaq, Norman Calder, George Makdisi, al-Mawardi

Ottoman Law: Halil İnalcık, Colin Imber, Uriel Heyd, Timur Kuran, Carter Findley

Venetian Law: Gaetano Cozzi, Stanley Chojnacki, Donald Queller, Frederic Lane, Diego Puga and Daniel Treffer

Mongol Law: David Morgan, Thomas Allsen, Paul Ratchnevsky, Igor de Rachewiltz, Valentin Riasanovsky, David Ayalon

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